

EXPERIMENT - 3

TO NEUTRALIZE SODIUM HYDROXIDE (NaOH) WITH HYDROCHLORIC ACID (HCl) SOLUTION AND TO OBTAIN CRYSTALS OF COMMON SALT.

APPARATUS REQUIRED

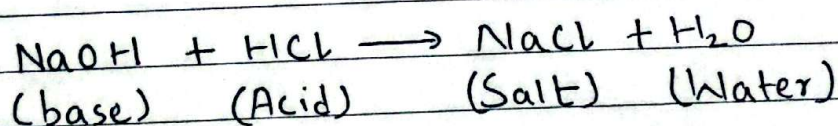
- | | |
|---------------------------|------------------------|
| a) <u>Porcelain basin</u> | b) <u>Test tubes</u> |
| c) <u>Beaker</u> | d) <u>Tripod stand</u> |
| e) <u>Funnel</u> | f) <u>Glass rod</u> |
| g) <u>Wire gauze</u> | h) _____ |

CHEMICAL REQUIRED

- | | |
|-------------------------------------|--------------------------------------|
| i. <u>Dilute NaOH solution</u> | ii. <u>Dilute HCl solution</u> |
| iii. <u>Silver nitrate solution</u> | iv. <u>Blue and red litmus paper</u> |

THEORY

When an acid is treated with a base, salt and water are produced. This reaction is known as acid-base reaction.



During acid-base reaction hydrogen ions of acid combine with hydroxide ions of base to form neutral product water and is also called neutralization reaction.

A substance that produces hydrogen ions (H^+) when dissolved in water is called acid.
e.g. - Hydrochloric acid (HCl)

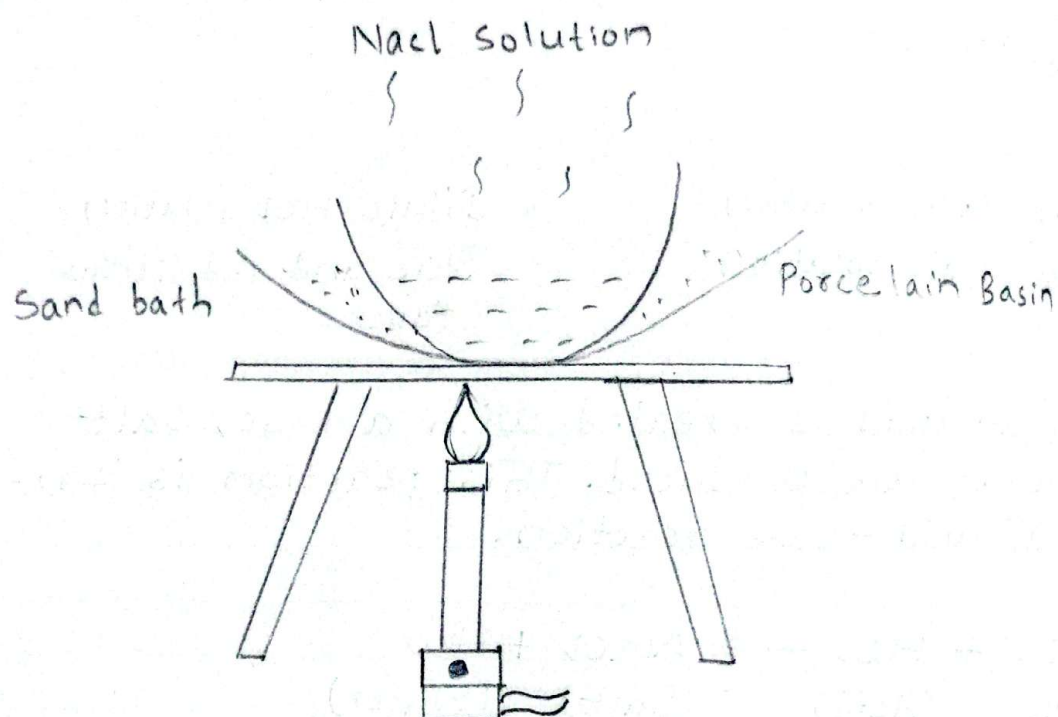


fig - Evaporation

A substance that produces hydroxide ions (OH^-) when dissolved in water is called base. e.g.: Sodium hydroxide - NaOH

A salt is a compound formed when the hydrogen ions of acid are replaced by metal ions.

An indicator is a substance that shows a visible change (such as color) to indicate whether a solution is acidic or basic. e.g. - litmus paper.

PROCEDURE

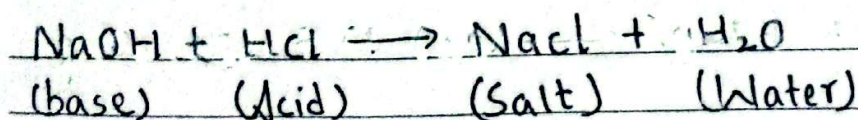
First, two drops of Hydrochloric acid (HCl) and two drops of sodium hydroxide (NaOH) were taken in a clean beaker and mixed well. Then, A drop of the mixture was then tested with red and blue litmus papers. The red litmus paper turned blue, showing that the solution was basic. More drops of HCl were then added slowly, mixed and tested again with litmus pap-

ers after each addition. When neither red nor blue litmus paper showed any color change, the solution was found to be neutral. The neutral solution was then kept for evaporation to dryness. After cooling, white crystals of sodium chloride (NaCl) were obtained. Finally, the crystals were shown to the teacher for observation and verification.

OBSERVATION TABLE

S.N.	Experiments	Observations	Inferences
1.	The solution was tested with red and blue litmus papers.	Blue litmus turned red or red litmus turned blue	The solution was acidic (blue-red) or basic (red-blue)
Few drops of NaOH were added to acidic solution and			
2.	Few drops of HCl were added to basic solution. The solution was tested again with red and blue litmus papers.	No change in colors was observed.	The solution became neutral.

TEST REACTION



RESULT

Common salt was obtained by the neutralization reaction of HCl and NaOH .

CONCLUSION

Hence, from above experiment it can be concluded that neutralization reaction can be carried out by mixing different types of acids and bases to prepare their salts.

PRECAUTIONS

- 1) Personal Protective equipment should be worn.
- 2) Equimolar solution of acid and base should be used for neutralization reaction.
- 3) Litmus paper should not be dipped in the solution as they may impart their colour to the solution and hence the salt becomes impure.

VIVA-VOCE

1. What is neutralization reaction?
2. What is crystal structure of NaCl?
3. What are the products in neutralization reaction of acetic acid (CH_3COOH) and caustic potash (KOH)?
4. Why is neutralization reaction of strong acid and strong base exothermic in nature?
5. Why is neutralization reaction known as double decomposition reaction or double displacement reaction?

1. Acid reacts with base to form salt and water or H^+ ion from acid and OH^- ion from a base to form H_2O . This reaction is called neutralization reaction.

2. Simple cubic crystal is crystal structure of NaCl.

3. Potassium acetate (CH_3COOK) and water (H_2O) are the products in neutralization reaction of acetic acid and caustic potash.

4. Since ΔH for this reaction is negative it is exothermic in nature.

In neutralization reaction of HCl and NaOH , 56 kJ/mol or 13.6 kcal heat energy released.



5. Because in neutralization reaction, positive ions and negative ions in two compounds switch partners to form two new compounds.